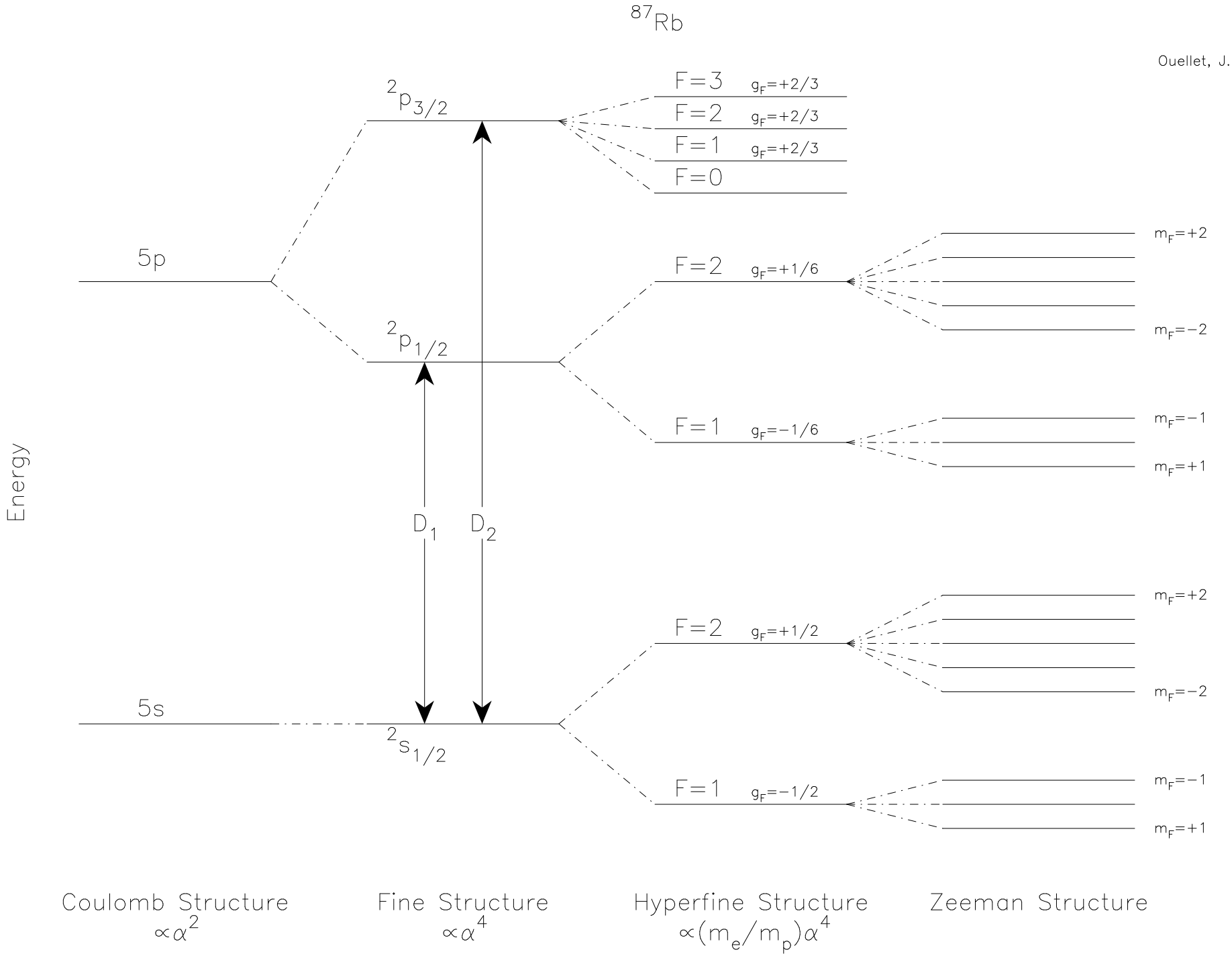
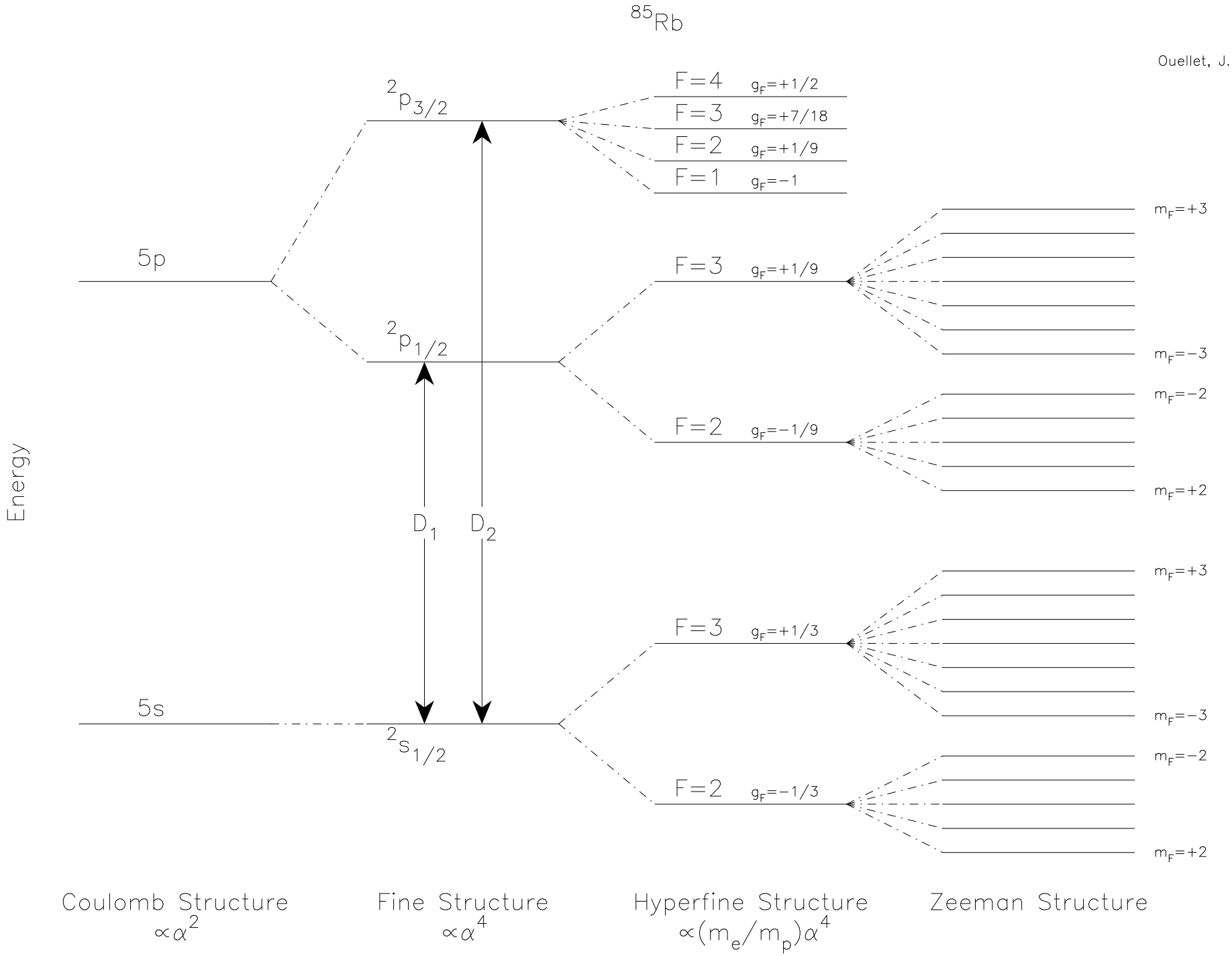
Energy Level Diagrams for Rb

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11/13/2010

Attached are the energy level diagrams for rubidium. These should not be cited as an academic resource, but used only as a reference.


^{87}Rb
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$^2p_{3/2}$
F=3 g_F=+2/3
F=2 g_F=+2/3
F=1 g_F=+2/3
F=0
5p
F=2 g_F=+1/6
m_F=+2
m_F=−2
$^2p_{1/2}$
F=1 g_F=−1/6
m_F=−1
m_F=+1
Energy
D_1
D_2
F=2 g_F=+1/2
m_F=+2
m_F=−2
5s
$^2s_{1/2}$
F=1 g_F=−1/2
m_F=−1
m_F=+1
Coulomb Structure
$\propto\alpha^2$
Fine Structure
$\propto\alpha^4$
Hyperfine Structure
$\propto(m_e/m_p)\alpha^4$
Zeeman Structure


^{85}Rb
Ouellet, J.
$^2p_{3/2}$
$^2p_{1/2}$
5p
5s
$^2s_{1/2}$
D_1
D_2
Energy
F=4 g_F=+1/2
F=3 g_F=+7/18
F=2 g_F=+1/9
F=1 g_F=−1
F=3 g_F=+1/9
F=2 g_F=−1/9
F=3 g_F=+1/3
F=2 g_F=−1/3
m_F=+3
m_F=−3
m_F=−2
m_F=+2
m_F=+3
m_F=−3
m_F=−2
m_F=+2
Coulomb Structure
$\propto\alpha^2$
Fine Structure
$\propto\alpha^4$
Hyperfine Structure
$\propto(m_e/m_p)\alpha^4$
Zeeman Structure

Zeeman Splitting of the $^2P_{3/2}$ States

Ouellet, J.

